

Reversals of age-related declines in neuronal signal transduction, cognitive, and motor behavioral deficits with blueberry, spinach, or strawberry ...

Ample research indicates that age-related neuronal-behavioral decrements are the result of oxidative stress that may be ameliorated by antioxidants. Our previous study had shown that rats given dietary supplements of fruit and vegetable extracts with high antioxidant activity for 8 months beginning at 6 months of age retarded age-related declines in neuronal and cognitive function. The present study showed that such supplements (strawberry, spinach, or blueberry at 14.8, 9.1, or 18.6 gm of dried aqueous extract per kilogram of diet, respectively) fed for 8 weeks to 19-month-old Fischer 344 rats were also effective in reversing age-related deficits in several neuronal and behavioral parameters including: oxotremorine enhancement of K(+)-evoked release of dopamine from striatal slices, carbachol-stimulated GTPase activity, striatal Ca(45) buffering in striatal synaptosomes, motor behavioral performance on the rod walking and accelerated tasks, and Morris water maze performance. These findings suggest that, in addition to their known beneficial effects on cancer and heart disease, phytochemicals present in antioxidant-rich foods may be beneficial in reversing the course of neuronal and behavioral aging.